Smart Equipment Care System (SECS)

A Web-Based University Equipment Issue, Maintenance, Budget Approval and Escalation Management System

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Objective

Centralized Platform

Develop a centralized web-based platform for managing equipment-related issues systematically, transparently, and efficiently

Easy Reporting

Enable easy reporting by students and teachers via equipment selection or QR code scanning

Technician Tools

Allow technicians to receive, diagnose, and manage issues with budget request capability

Authority Oversight

Provide higher authorities with a portal to review and approve/reject budget requests

Auto-Escalation

Automatically escalate unresolved issues when SLA is exceeded

Digital History

Maintain complete digital history and improve accountability and communication



The Problem & Our Solution

The Challenge

University equipment failures are reported through informal channels — verbal complaints, phone calls, or paper records — leading to lost tickets, unclear accountability, and delayed repairs.

How SECS Solves It

SECS provides a centralized, role-based platform where issues are reported digitally, tracked transparently, and escalated automatically when SLA deadlines are exceeded.

System Scope: Three User Portals



Student / Teacher Portal

- Register and view equipment
- Scan QR codes to submit issues
- Track progress and receive notifications



Technician Portal

- Accept or reject reported issues
- Diagnose problems and request budgets
- Update progress and mark resolved



Higher Authority Portal

- Review issues and monitor escalations
- Approve or reject budget requests
- Assign support and track resolution

Equipment Management & Escalation



Equipment Database

Maintain equipment database with ID, name, category, location, department, status, and unique QR code.

SLA-Based Escalation Tiers

Automatic escalation to higher authority when resolution deadline is exceeded.

🔴 **Critical**
24 hours

🟠 **High**
3 days

🟡 **Medium**
7 days

🟢 **Low**
14 days

Feasibility Assessment



Technical

HTML, CSS, Bootstrap, JavaScript (frontend); PHP/Laravel (backend); MySQL database — widely used, suitable technologies



Economic

Low cost using free/open-source tools; main expenses are optional hosting and domain



Operational

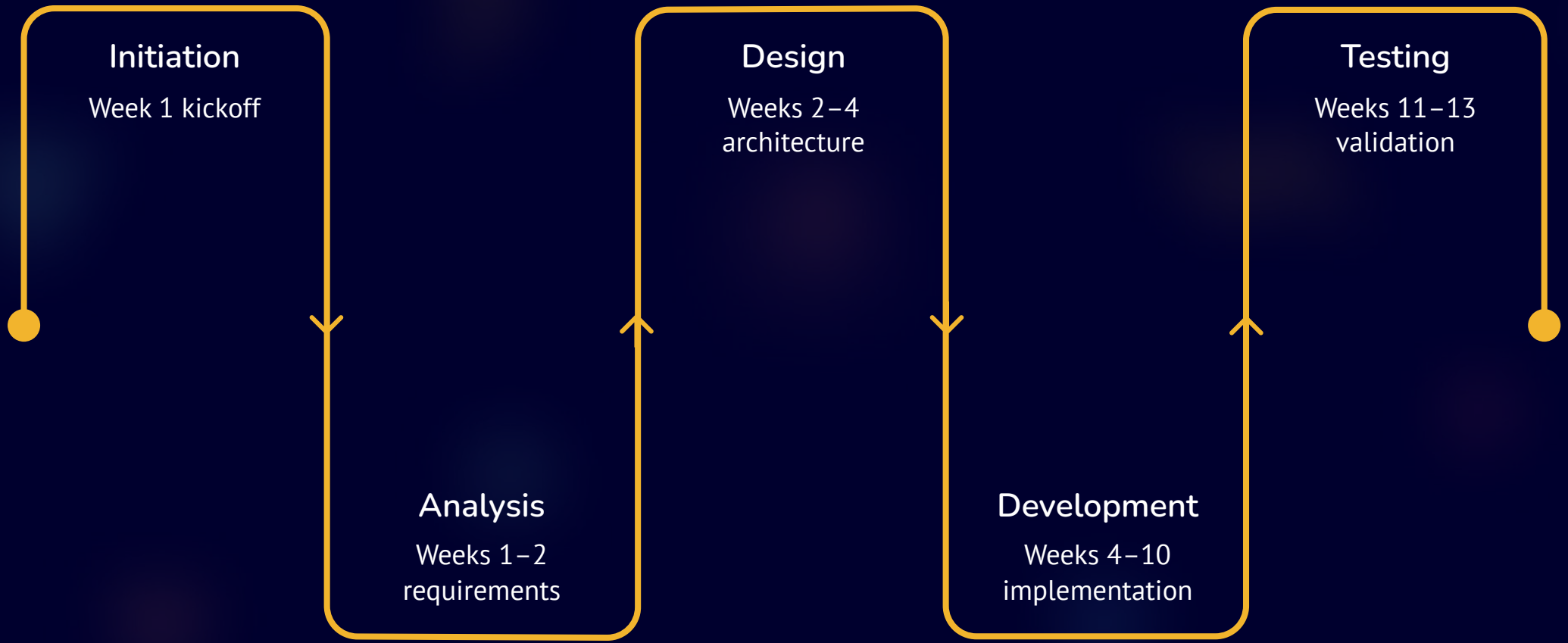
Replaces informal communication with centralized system; improves transparency and accountability



Schedule

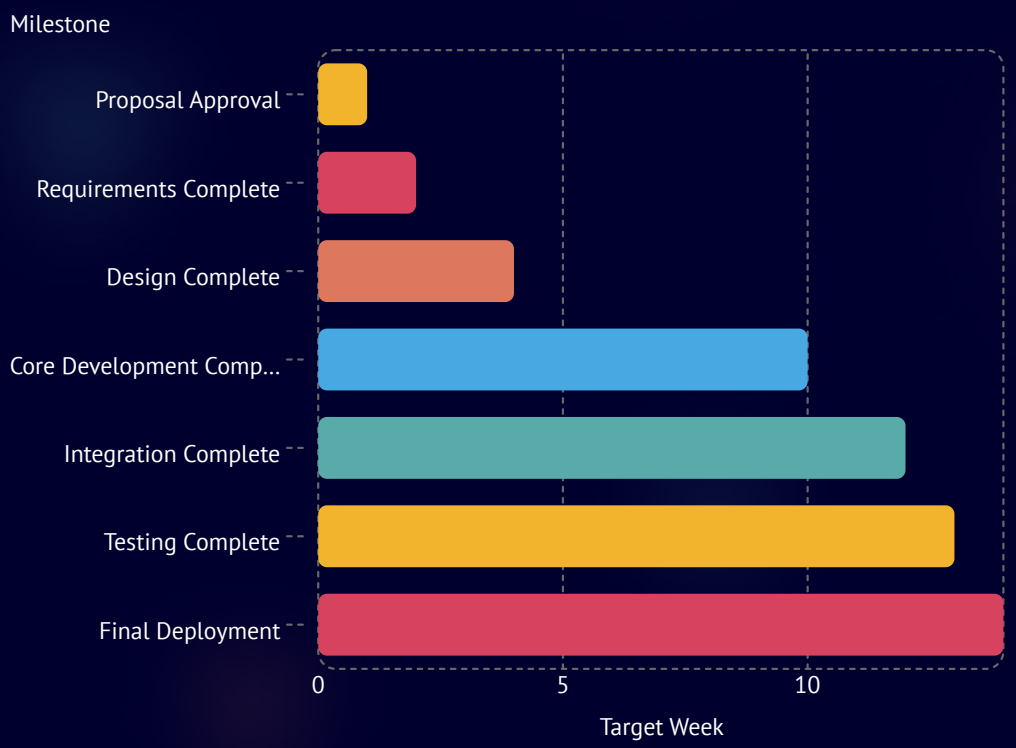
Feasible within 14-week academic timeline

Work Breakdown Structure



The project follows a structured 14-week lifecycle from initiation through final documentation and presentation, with development spanning the largest portion of the timeline.

Project Timeline & Milestones



Duration: 14 Weeks

Parallel development of modules across the timeline:

- Student/Teacher Portal
- Technician Portal
- Authority Portal
- Equipment Management
- Issue Management
- Budget Module
- SLA / Escalation Engine
- Notifications System

Motivation

University equipment issues are often handled through **manual and unorganized communication**.

Students may not know **where or whom to report** an issue.

Repair progress and issue status are difficult to **track**.

Budget requests and approvals can cause **delays**.

Unresolved issues may remain unnoticed without **automatic escalation**.

Therefore, we are motivated to develop **SECS** to make equipment maintenance **faster, transparent, and accountable**.

Budget & Conclusion

6,500

BDT Total Cost

Primarily optional hosting and documentation expenses

Key Assumptions

- Free/open-source tools throughout
- Local development server
- No developer salaries

Conclusion

SECS is a practical solution that reduces communication gaps, improves accountability, and is technically, economically, operationally, and schedule feasible for academic implementation.